

Section 1 Introduction to Statistics

Statistics I · Fall 2026 · Zong-Wei Yeh · Department of Finance, CCU

Overview

Almost every important decision is made without complete information. A fund manager picks a portfolio before knowing next year's returns. A firm launches a product without asking every possible buyer. Both can still collect some data. But the data cover only part of the group the manager and the firm want to understand. That part is a sample. The decision must then be made from the sample.

Statistics is the subject that makes that step safe. It does two things. First, it shows how to summarize what we observed. Second, it tells us how far to trust that summary. The second thing is the harder one. We have seen only a part, and we want to speak about the whole.

This section builds the vocabulary used in the rest of the course. It has three parts.

- §1.1 splits statistics into two kinds. One kind describes the data we have. The other kind draws conclusions about data we do not have.
- §1.2 defines four basic terms: population, sample, parameter, statistic. It also defines how we draw a sample. It then shows how a data set can mislead us, even when the data set is very large.
- §1.3 sorts data into types. It then asks which summaries each type allows.

None of this is hard. But every symbol in Sections 2–7 goes back to one of these ideas.

1.1 What Is Statistics?

Raw *data* are recorded observations. Examples are 250 daily closing prices, the answers of 800 survey respondents, and the defect counts of 40 batches. Data alone tell us nothing. They become *information* when we organize them so that a pattern appears. That information helps *decision-making* only after we judge how reliable the pattern is. Statistics does both of those steps. It works in two modes. Section 2 covers the first mode. Sections 3–7 build the second.

Definition 1.1 (Descriptive and inferential statistics).

- (i) Descriptive statistics (敘述統計): *methods, in graphs and in numbers, that organize and summarize a set of data.*
- (ii) Inferential statistics (推論統計): *methods that use data from a sample to draw conclusions about the larger population it came from. The conclusions may be estimates, tests or forecasts.*

The two modes use almost the same arithmetic. They differ in how much the conclusion claims.

Suppose you report the average return of the 30 stocks you hold. That is a _____ statement. It is a fact about those 30 stocks, and it is exactly right. Now suppose you draw 30

stocks at random from a market of 900. You use their average return to speak about the whole market. That is an _____ statement. It is a claim about 900 stocks based on 30. So it can be wrong.

This kind of statement always carries uncertainty. This course exists to measure that uncertainty, not to hide it.

To measure it we need a probability (機率) model. The model says how the sample was produced. We build it in Sections 3–6. First we need names for the objects the model connects. §1.2 defines them, and we come back to the whole picture there.

1.2 Population and Sample

Definition 1.2 (Population and sample).

- (i) The population (母體) is the _____ set of items of interest, of size N (possibly infinite).
- (ii) A sample (樣本) is an _____ of the population, of size n .

The *question* decides the population. The data you already have do not decide it. Suppose you want to know how Taiwanese households invest. Then the population is all Taiwanese households. One brokerage's clients are not that population. Interviewing more of them does not fix this. People who open a brokerage account are already a special group. Households with no account never reach the list at all.

Now suppose we want a number that describes the whole population. One example is the average holding of all Taiwanese households. Only one method gives that number for certain. We measure every member of the population. This is a *census* (普查).

A census is usually not possible. There are three common reasons.

- (1) **Cost and time.** The government runs Taiwan's population and housing census once every ten years. No firm can repeat that to answer a question this month.
- (2) **Measuring destroys the item.** To learn how long a light bulb lasts, we must run it until it fails. If we test every bulb, we have no bulbs left to sell.
- (3) **The population cannot be listed at all.** This case matters most, and it has its own name.

A population does not have to be a set of things that already exist. Suppose we want the average daily return of one stock. To know this number for certain, we would need a census. That census is every daily return the stock will ever have. We may hold 250 trading days of data. We may hold ten years. Neither one is the full list. The stock is still trading, so more returns are still coming.

There is a second reason too, and it is the deeper one. The returns we saw are only the ones that did happen. We want to know what the stock *could* return, not only what it *did* return. So the population is every return the stock could produce. A population like this is called _____. It has no list, so no census can reach it.

Note carefully what the problem is here. Ten years of returns is not a bad sample. It is simply not the whole population.

So we sample. We measure a part, and we reason about the whole. Sampling is not a worse choice than a census. For most questions it is the only choice there is. The rest of this course shows how to use it honestly.

We now have two kinds of numbers to keep apart. One describes the population, and it is the one we want. The other is computed from the sample, and it is the one we can actually get. Confusing them is the most common beginner's error.

Definition 1.3 (Parameter and statistic).

- (i) A parameter (母數/參數) is a numerical measure of a population characteristic. It is _____, and usually unknown.
- (ii) A statistic (統計量) is a numerical measure computed from a sample. We know its value once we have the sample, but it _____.

The difference here matters, so read it slowly. The parameter is the mean return μ of all 900 stocks. Its value never changes, and we do not know it. The statistic is the mean $\bar{x}$ of your 30 stocks. We know it exactly. But it is not the same for every sample. Another 30 stocks give another $\bar{x}$. So the number we *want* never changes, and it is hidden. The number we *have* is visible and random. Section 7 treats a statistic as a random variable and studies its distribution.

We now have all four objects, so we can draw the whole picture. Figure 1 is the logic of the course. We fix a population. We draw a sample by a procedure we choose. We compute a statistic. Then we reason back to the parameter. The forward step is chance. The backward step is inference. Probability measures the gap between them.

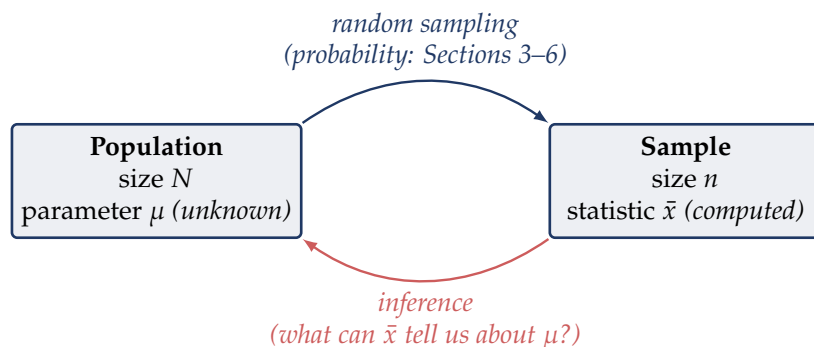


Figure 1: The inference loop. We can model the forward arrow, because we know how the sample was drawn. Probability then tells us how $\bar{x}$ behaves. The backward arrow is the one we want. It is only as good as our model of the forward arrow. A sample statistic is usually not equal to the parameter. We can never check whether our sample gave the exact value. So the honest question is not “is it right?” It is “how far off can it be?”

We can trust the forward arrow only when we know how the sample was drawn.

Definition 1.4 (Simple random sample).

A simple random sample (簡單隨機樣本) of size n is one selected so that

- (i) every member of the population is _____ to be chosen;
- (ii) choosing one member does not change the chance of choosing another; and
- (iii) every possible sample of size n has the same chance of being chosen.

It is usually just called a random sample (隨機樣本).

We add one note here, because we need it later. In practice we draw *without replacement* (取後不放回) from a finite population. Then the draws are not strictly independent. Once we take a member, it cannot appear again. So condition (ii) holds only approximately. When n is small compared with N , say n/N below 5%, this effect is very small. We then treat the observations as independent. Nearly every formula in Sections 3–7 uses that assumption. Section 5 gives the one model that does not. It handles drawing without replacement directly. Section 3 defines independence precisely.

The definition controls the _____, not the result. Here is what that means. Suppose we draw 30 stocks at random from the 900. By bad luck they could all turn out to be large firms. Their mean return $\bar{x}$ would then sit far from μ . We did nothing wrong. We followed the rules exactly, and this one draw was unlucky.

But we know that every sample of 30 had the same chance of being drawn. So we can compute how often such bad luck happens. This is the trade-off. Random sampling does not promise a representative sample. It promises that we can put a number on the chance of a bad one.

A convenience sample (便利樣本) is different. Suppose we judge the whole market from the stocks on the front page of Yahoo Finance. We took those stocks because they were easy to find. Nobody tells us why the site put those stocks there. So we do not know any stock's chance of appearing on that page. Here we cannot put a number on anything.

A common practical alternative is *systematic sampling* (系統抽樣). Order the list. Pick a random start among the first j items. Then take every j th item. The sampling interval (抽樣間隔) is

$$j = \quad .$$

This is cheap, and it usually behaves like a random sample. But that holds only if the ordering has nothing to do with the variable we study. The danger is a _____ pattern in the list with period j , or with a period that divides j . Every selected item then falls at the same place in that repeating pattern. The sample is then not like the population.

Example 1.5. A brokerage's client list holds $N = 8,000$ accounts, ordered by the date each was opened. An analyst wants a systematic sample of $n = 200$.

- (a) Find the sampling interval j .
- (b) With random start 7, give the first three accounts selected and the last one.
- (c) Describe one ordering of the list that would make the sample misleading.

Even when we sample perfectly, the number we get usually differs from the parameter. That gap has two sources. They have separate names, because only one of them is a statistics problem.

Sampling error (抽樣誤差) comes from observing n items rather than all N . It is unavoidable, and it is not a mistake. It is also _____. We know how the sample was drawn, so we can say how large the error is likely to be. It also shrinks as n grows. Sections 3–7 study it in detail.

Nonsampling error (非抽樣誤差) is everything that would remain even in a complete _____. Three sources are common:

- (1) **Wrong frame.** Two populations are in play here. The *target population* is the one the question is about. The *sampled population* is the one your procedure can actually reach. The sampling frame (抽樣底冊) you draw from is the list of the second one. If that list leaves out part of the target, the two populations differ. Landline numbers, your own customers, and the funds that still exist are all examples.
- (2) **Inaccurate response.** The question may be unclear. It may push the respondent toward one answer. Or it may be too private for an honest answer. "How much have you stolen from this plant?" will not give reliable data.
- (3) _____. People who refuse are rarely a random subset. You end up with a random sample of the population *willing to respond*.

So there is a warning here. Sampling error shrinks toward zero as the sample grows. The *bias* (偏誤) from a nonsampling error does not. A larger sample makes the sampling error smaller. But that error is now measured from the wrong number. The bad procedure still measures the wrong thing. A biased survey of a million people is no better than one of a thousand.

The three sources above ask where the error comes from. We now ask a different question. Which population does the number actually describe? Here the problem is one that no larger sample can repair. It takes three forms, and all three are nonsampling errors. The question is which items entered the data set, not how many.

Selection bias (選擇偏誤) happens when an item enters the sample for a reason connected to _____. The sampled population then differs from the target population in a systematic way.

The first form is a wrong frame. We met it above under that name. Note what changed. There we asked where the error comes from. Here we ask what the number is a fact about. The two questions have different answers, so they get different names.

Suppose a brokerage wants the average annual return of *all* retail investors in the country. It estimates that return from the accounts in its premium research service. Subscribers trade more than the typical investor. They also hold larger accounts. So their returns are different in a systematic way. The frame the brokerage sampled from is its subscriber list, and that is not the target population. The number is a fact about subscribers, not about all retail investors. The sample can be huge, and the bias does not go away.

The second form is the everyday one. We call it _____. Online reviews are written by customers who chose to write one. So the average rating describes only customers with strong opinions. It does not describe all customers. The same problem appears in finance whenever reporting is voluntary. A hedge fund index built from funds that choose to report describes the funds *willing to report*. It does not describe “hedge funds”.

The third form comes from time. This one matters most for the return data you meet in finance. A fund database holds the funds that *still exist*. Funds that did badly were closed or merged away, and their returns went with them. So the funds left in the database are the ones that did well. So the average describes a group we picked because it did well. We picked it only after we saw the result. The data show no sign of this. We call it _____ bias.

Example 1.6. Ten years ago a database tracked 100 equity funds. Today 70 still operate. The other 30 were closed or merged away during the decade. Over the ten years those 70 averaged 8.0% per year. The 30 that disappeared averaged 2.0% per year while they existed. A magazine uses today's database and reports: "the average equity fund returned 8.0% a year over the decade."

- (a) Average over all 100 funds the database began with. Weight each fund equally, and let each contribute its own average annual return. What is the mean annual return of all 100 funds?
- (b) By how much does the magazine's figure overstate it?
- (c) Which population does 8.0% really describe, and why is it useless to an investor choosing a fund today?

1.3 Data Types and Measurement Levels

Before you choose any method, you must know what kind of data you have. The kind of data decides which summaries mean anything at all.

Definition 1.7 (Variable; categorical and numerical).

A variable is a characteristic that may differ across the population. It is one of two types:

- (i) **categorical** (類別變數): *the values are groups or labels, such as industry, gender, or a 1–5 satisfaction rating. This holds even when we store them as numeric codes.*
- (ii) **numerical** (數值變數): *the values are quantities, found by counting or by measuring. A numerical variable is discrete (離散) if we can list its values one by one. It is continuous (連續) if it can take any value in an interval.*

So the number of trades a client places in a month is _____. The market value of their portfolio is _____. There are two common traps here.

- **Codes are not quantities.** A satisfaction rating stored as 1, . . . , 5 has numeric entries. But "5" is only a label for "very satisfied". We picked that label ourselves, and "A" would work just

as well. A rating stored as a number is still a _____ variable.

- **Rounding is not the same as counting.** We record a share price to two decimals. The record then shows only values with two decimals. But the price itself can take any value in a range. We classify the quantity, not the way we wrote it down.

A second classification asks which arithmetic the numbers allow. It comes from Stevens (1946). It has four measurement levels (衡量尺度).

Definition 1.8 (Measurement levels).

- (i) Nominal (名目尺度): *labels with no order.*
- (ii) Ordinal (順序尺度): *we can _____ the values, but the differences carry no fixed meaning.*
- (iii) Interval (等距尺度): *the differences are meaningful, but the zero point is _____.*
- (iv) Ratio (比率尺度): *the differences are meaningful and the zero is _____. So ratios of values make sense.*

Nominal and ordinal data are qualitative (質性資料/ 定性資料); interval and ratio data are quantitative (量化資料/ 定量資料).

The two classifications fit together. Nominal and ordinal are the levels of a variable whose values are labels. Interval and ratio are the levels of _____ ones. So classifying a variable takes two steps. First decide which of the two types it is. Then decide which of its two levels. Most mistakes happen at the last two levels. The test is simple. Ask whether the zero is a _____, or whether zero means there is none of the thing. Someone chose the zero on the Celsius scale, so 20°C is not twice as warm as 10°C. Zero dollars means no money at all, so NT\$2m really is twice NT\$1m.

Level	Supports	Does not support	Example
Nominal	counting, mode	ordering, differences	industry sector
Ordinal	ordering, median	differences, means	bond rating (AAA, AA, ...)
Interval	differences, mean	ratios	temperature (°C), calendar year
Ratio	differences, ratios	—	portfolio value, trading volume

Table 1: The four measurement levels and the arithmetic each one allows. Read the table *downward*. Each row supports everything the row above supports, and one more kind of arithmetic. The important column is the third one. It tells you which computations are *meaningless*.

This matters most with ordinal data coded as numbers. A satisfaction scale runs from 1 (very dissatisfied) to 5 (very satisfied). You can compute the average of those numerals. But the gap between “4” and “5” need not equal the gap between “1” and “2”. The codes carry order, not _____. So the mean of such codes has no clear meaning. The median and the frequency distribution are the safe summaries. The mean assumes equal gaps. The scale does not give equal gaps. Section 2 defines the mode, the median and the frequency distribution.

Example 1.9. A bank stores three fields for each of its 500 corporate clients:

- (i) the client's credit rating (AAA, AA, ..., C);
- (ii) the calendar year in which the account was opened;
- (iii) the balance on the account, in NT\$.

For each computation below, say whether it is meaningful, and why.

- (a) The mean credit rating.
- (b) The difference between two clients' opening years.
- (c) The ratio of two clients' balances.
- (d) The ratio of two clients' opening years.

Exercises

Exercise 1.1. A bank surveys a random sample of its credit-card holders. For each variable, state whether it is categorical or numerical; if categorical give the measurement level, if numerical say discrete or continuous.

- (a) The type of card held (Visa, Mastercard, JCB).
- (b) Satisfaction with the mobile app (1: very dissatisfied, ..., 5: very satisfied).
- (c) The number of late payments made last year.
- (d) The amount charged to the card last month, in NT\$.

Exercise 1.2. A researcher studies the 1,200 firms listed on the Taiwan Stock Exchange. She draws 60 at random and finds their mean return on equity is 9.4%; the mean ROE of all 1,200 firms is unknown.

- (a) State the population and its size N .
- (b) State the sample and its size n .
- (c) Is 9.4% a parameter or a statistic?
- (d) Suppose she drew a different 60 firms. Which number would change: the mean ROE of the 1,200, or the mean of her sample? What does this say about which one is random?

Exercise 1.3. An auditor must examine $n = 120$ invoices from a file of $N = 3,600$, arranged in the order they were issued, and samples systematically.

- (a) Find the sampling interval j .
- (b) With random start 17, give the first three invoices selected and the last one.
- (c) The firm issues invoices in a fixed daily cycle: one per branch, for its 30 branches, always in the same order. What goes wrong, and how would you fix it?

Exercise 1.4. Classify each as sampling error or nonsampling error, with a one-sentence reason.

- (a) The mean income of the 500 sampled households differs from the mean income of all households in the city.
- (b) Forty percent of the households contacted refuse the income question.
- (c) The sampling frame is a list of landline telephone numbers.
- (d) Respondents systematically understate their income.

In which of these is the bias reduced by contacting 5,000 households instead of 500?

Exercise 1.5. A data vendor's database began five years ago with 250 hedge funds. Today 200 remain, averaging 7.5% per year over the five years; the 50 that closed averaged 1.5% per year while they operated.

- (a) What mean annual return does the vendor report using only the funds now in the database?

- (b) *Average over all 250 original funds. Weight each fund equally, and let each contribute its own average annual return. What is the mean annual return of all 250 funds?*
- (c) *State the size and direction of the bias.*
- (d) *Would tracking 1,000 surviving funds instead of 200 give a better figure? Explain.*

Exercise 1.6. *An online retailer shows a mean customer rating of 4.6/5, computed from the reviews it received. Separately, a hedge fund index is computed each month from the returns of funds that voluntarily report to the provider.*

- (a) *Name the bias in each case, and state precisely which population each number describes.*
- (b) *The 4.6 is an average of an ordinal variable. Explain why that is a second, separate problem.*
- (c) *Propose one concrete change to the retailer's data collection that removes the bias in (a), and say honestly what it costs and what remains.*